

## Consequences of Overfitting the van der Waals Radii of Ions

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Cite This: *J. Chem. Theory Comput.* 2023, 19, 2064–2074

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**ABSTRACT:** Atomic radii play important roles in scientific research. The covalent radii of atoms, ionic radii of ions, and van der Waals (VDW) radii of neutral atoms can all be derived from crystal structures. However, the VDW radii of ions are a challenge to determine because the atomic distances in crystal structures were determined by a combination of VDW interactions and electrostatic interactions, making it unclear how to define the VDW sphere of ions in such an environment. In the present study, we found that VDW radii, which were determined based on the 0.0015 au electron density contour through a wavefunction analysis on atoms, have excellent agreement with the VDW radii of noble-gas atoms determined experimentally. Based on this criterion, we calculated the VDW radii for various atomic ions across the periodic table, providing a systematic set of VDW radii of ions. Previously we have shown that the 12-6 Lennard-Jones nonbonded model could not simultaneously reproduce the hydration free energy (HFE) and ion–oxygen distance (IOD) for an atomic ion when its charge is +2 or higher. Because of this, we developed the 12-6-4 model to reproduce both properties at the same time by explicitly considering the ion-induced dipole interactions. However, recent studies showed it was possible to use the 12-6 model to simulate both properties simultaneously when an ion has the  $R_{\min}/2$  parameter (i.e., the VDW radius) close to the Shannon ionic radius. In the present study, we show that such a “success” is due to an unphysical overfitting, as the VDW radius of an ion should be significantly larger than its ionic radius. Through molecular dynamics simulations, we show that such overfitting causes significant issues when transferring the parameters from ion–water systems to ion–ligand and metalloprotein systems. In comparison, the 12-6-4 model shows significant improvement in comparison to the overfitted 12-6 model, showing excellent transferability across different systems. In summary, although both the 12-6-4 and 12-6 models could reproduce HFE and IOD for an ion, the 12-6-4 model accomplishes such a task based on the consideration of the physics involved, while the 12-6 model accomplishes this through overfitting, which brings significant transferability issues when simulating other systems. Hence, we strongly recommend the use of the 12-6-4 model (or even more sophisticated models) instead of overfitted 12-6 models when simulating complex systems such as metalloproteins.

|             |    |                         |                         |                         |                         |                         |    |                          |                          |                          |                          |                         |                         |                          |    |                         |                          |                          |                          |                         |                          |                          |                          |                           |                            |                            |                            |                            |                            |                            |                            |                            |    |                          |                          |                         |                          |                          |                          |                          |                          |                           |                            |                            |                            |                            |                            |                            |                            |                            |                            |    |                          |                          |                          |                          |                          |                         |                          |                          |                           |                            |                            |                            |                            |                            |                            |                            |                            |                            |                            |                            |                           |                            |                            |                            |                            |                            |                            |                            |                            |                            |                            |                            |    |                          |                          |                          |                          |                          |                         |                          |                          |                           |                            |                            |                            |                            |                            |                            |                            |                            |                            |                            |                            |                           |                            |                            |                            |                            |                            |                            |                            |                            |                            |                            |                            |    |
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| H<br>1:1.67 | He | B<br>+3:0.63<br>+4:0.53 | C<br>+2:0.50<br>+3:0.40 | N<br>+3:0.40<br>+4:0.30 | O<br>+3:0.30<br>+4:0.20 | F<br>+3:0.20<br>+4:0.10 | Ne | Na<br>+1:1.02<br>+2:1.12 | Mg<br>+2:1.02<br>+3:1.12 | Al<br>+3:1.02<br>+4:1.12 | Si<br>+4:1.02<br>+5:1.12 | P<br>+5:1.02<br>+6:1.12 | S<br>+6:1.02<br>+7:1.12 | Cl<br>+7:1.02<br>+8:1.12 | Ar | K<br>+1:1.67<br>+2:1.12 | Ca<br>+2:1.12<br>+3:1.12 | Sc<br>+3:1.12<br>+4:1.12 | Ti<br>+4:1.12<br>+5:1.12 | V<br>+5:1.12<br>+6:1.12 | Cr<br>+6:1.12<br>+7:1.12 | Mn<br>+7:1.12<br>+8:1.12 | Fe<br>+8:1.12<br>+9:1.12 | Co<br>+9:1.12<br>+10:1.12 | Ni<br>+10:1.12<br>+11:1.12 | Cu<br>+11:1.12<br>+12:1.12 | Zn<br>+12:1.12<br>+13:1.12 | Ga<br>+13:1.12<br>+14:1.12 | Ge<br>+14:1.12<br>+15:1.12 | As<br>+15:1.12<br>+16:1.12 | Se<br>+16:1.12<br>+17:1.12 | Br<br>+17:1.12<br>+18:1.12 | Kr | Rb<br>+1:1.67<br>+2:1.12 | Sr<br>+2:1.12<br>+3:1.12 | Y<br>+3:1.12<br>+4:1.12 | Zr<br>+4:1.12<br>+5:1.12 | Nb<br>+5:1.12<br>+6:1.12 | Mo<br>+6:1.12<br>+7:1.12 | Tc<br>+7:1.12<br>+8:1.12 | Ru<br>+8:1.12<br>+9:1.12 | Rh<br>+9:1.12<br>+10:1.12 | Pd<br>+10:1.12<br>+11:1.12 | Ag<br>+11:1.12<br>+12:1.12 | Cd<br>+12:1.12<br>+13:1.12 | In<br>+13:1.12<br>+14:1.12 | Sn<br>+14:1.12<br>+15:1.12 | Pb<br>+15:1.12<br>+16:1.12 | Bi<br>+16:1.12<br>+17:1.12 | Po<br>+17:1.12<br>+18:1.12 | At<br>+18:1.12<br>+19:1.12 | Rn | Fr<br>+1:1.67<br>+2:1.12 | Ra<br>+2:1.12<br>+3:1.12 | Ac<br>+3:1.12<br>+4:1.12 | Th<br>+4:1.12<br>+5:1.12 | Pa<br>+5:1.12<br>+6:1.12 | U<br>+6:1.12<br>+7:1.12 | Np<br>+7:1.12<br>+8:1.12 | Pu<br>+8:1.12<br>+9:1.12 | Am<br>+9:1.12<br>+10:1.12 | Cm<br>+10:1.12<br>+11:1.12 | Bk<br>+11:1.12<br>+12:1.12 | Cf<br>+12:1.12<br>+13:1.12 | Es<br>+13:1.12<br>+14:1.12 | Fm<br>+14:1.12<br>+15:1.12 | Md<br>+15:1.12<br>+16:1.12 | No<br>+16:1.12<br>+17:1.12 | Lr<br>+17:1.12<br>+18:1.12 | Lu<br>+18:1.12<br>+19:1.12 | Hf<br>+19:1.12<br>+20:1.12 | Ta<br>+20:1.12<br>+21:1.12 | W<br>+21:1.12<br>+22:1.12 | Re<br>+22:1.12<br>+23:1.12 | Os<br>+23:1.12<br>+24:1.12 | Ir<br>+24:1.12<br>+25:1.12 | Pt<br>+25:1.12<br>+26:1.12 | Au<br>+26:1.12<br>+27:1.12 | Hg<br>+27:1.12<br>+28:1.12 | Tl<br>+28:1.12<br>+29:1.12 | Pb<br>+29:1.12<br>+30:1.12 | Bi<br>+30:1.12<br>+31:1.12 | Po<br>+31:1.12<br>+32:1.12 | At<br>+32:1.12<br>+33:1.12 | Rn | Fr<br>+1:1.67<br>+2:1.12 | Ra<br>+2:1.12<br>+3:1.12 | Ac<br>+3:1.12<br>+4:1.12 | Th<br>+4:1.12<br>+5:1.12 | Pa<br>+5:1.12<br>+6:1.12 | U<br>+6:1.12<br>+7:1.12 | Np<br>+7:1.12<br>+8:1.12 | Pu<br>+8:1.12<br>+9:1.12 | Am<br>+9:1.12<br>+10:1.12 | Cm<br>+10:1.12<br>+11:1.12 | Bk<br>+11:1.12<br>+12:1.12 | Cf<br>+12:1.12<br>+13:1.12 | Es<br>+13:1.12<br>+14:1.12 | Fm<br>+14:1.12<br>+15:1.12 | Md<br>+15:1.12<br>+16:1.12 | No<br>+16:1.12<br>+17:1.12 | Lr<br>+17:1.12<br>+18:1.12 | Lu<br>+18:1.12<br>+19:1.12 | Hf<br>+19:1.12<br>+20:1.12 | Ta<br>+20:1.12<br>+21:1.12 | W<br>+21:1.12<br>+22:1.12 | Re<br>+22:1.12<br>+23:1.12 | Os<br>+23:1.12<br>+24:1.12 | Ir<br>+24:1.12<br>+25:1.12 | Pt<br>+25:1.12<br>+26:1.12 | Au<br>+26:1.12<br>+27:1.12 | Hg<br>+27:1.12<br>+28:1.12 | Tl<br>+28:1.12<br>+29:1.12 | Pb<br>+29:1.12<br>+30:1.12 | Bi<br>+30:1.12<br>+31:1.12 | Po<br>+31:1.12<br>+32:1.12 | At<br>+32:1.12<br>+33:1.12 | Rn |
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## INTRODUCTION

Atomic radii are a fundamental concept in chemistry. Different atomic radii have been defined, including covalent radii,<sup>1</sup> ionic radii,<sup>2–4</sup> metallic radii,<sup>5</sup> and van der Waals (VDW) radii.<sup>6,7</sup> In these radii, the atoms are usually represented by spheres. Although atoms can display anisotropy while forming a chemical bond, the spherical approximation is still the most widely used approximation due to its simplicity. The concept of atomic radii has been employed extensively in scientific research. They are usually used to determine the chemical bonds between atoms and the sizes of molecules.<sup>8,9</sup> For example, covalent radii and metallic radii can be used to determine the covalent and metallic bonds and provide equilibrium bond lengths for molecular simulations involving such bonds. The VDW volume describes the volume occupied by a molecule. Likewise, the VDW radius is the contact distance between atoms/molecules.<sup>10</sup> VDW radii have been used in various applications, such as determining hydrogen bonding,<sup>11</sup> calculating the hydration free energy,<sup>12</sup> as well as in

crystal chemistry and stereochemistry.<sup>13</sup> Furthermore, a reasonable set of VDW radii is quite important for one to get reasonable atomic distances when modeling molecular systems in computational chemistry. VDW radii are also needed in the ESP<sup>14</sup> and CHPLPG<sup>15</sup> fitting schemes for choosing the spatial extent of the electrostatic potential to fit atomic partial charges. Because of this importance, we believe it is critical to have physically meaningful parameters for the VDW radii in molecular modeling.

The covalent radii are for atoms having covalent bonds with each other.<sup>1</sup> One can determine whether there is a covalent bond between two atoms by comparing their atomic distance

Received: December 9, 2022

Published: March 23, 2023



with the sum of their covalent radii (usually within a certain tolerance). Metallic radii are defined similarly but for metals forming metallic bonds, which is intrinsically different from covalent bonds.<sup>2</sup> The ionic radii are defined for ions in ionic crystal structures. The sum of ionic radii is used to determine the atomic distance between the neighboring counterions in the crystal lattice.<sup>3</sup> These three different radii are used for covalent, metallic, and ionic bonds. One can say that the covalent interaction determines the covalent bonds and metallic interactions determine the metallic bonds, with both the covalent and metallic bonds can be termed as the bonded interactions in molecular modeling. However, the ionic radii are not determined by only the VDW or electrostatic interactions, but by the sum of them, which is termed as the nonbonded interactions in molecular modeling. In ionic crystal structures, the balance between VDW and electrostatic interactions is so important that without the VDW interactions, ions with opposite charges will collapse because of the attractive force. Similarly, without the electrostatic attraction, counterions will have much larger distances from each other.<sup>16</sup>

The same issue exists for the VDW radii of ions. Specifically, it is quite challenging to determine the VDW radii of ions based on crystal structures because the electrostatic and VDW interactions coexist. The nonbonded interactions have both components by nature, where their decomposition is difficult for systems having atoms with partial charges. Fortunately, systems that only have VDW interactions exist, which are the noble-gas atoms. Scientists determined their radii experimentally,<sup>12</sup> and these radii are intrinsically VDW radii. However, for atoms other than noble-gas atoms, experimental determination of their VDW radii is not straightforward and usually needs extra assumptions. Inconsistencies in assumptions and methods among different studies lead to a range of VDW radii. For example, one may assume atoms are spheres<sup>6,7</sup> and another may not but include the anisotropic effect.<sup>17</sup> Oxygen atoms could be used as a probe, while nitrogen or other elements may also be used.<sup>18</sup>

Besides experimental approaches, theoretical calculation is another way to determine the VDW radii of atoms and ions. For example, Stokes calculated the VDW radii for 16 cations and anions which shared the same electronic structures as the noble-gas atoms.<sup>12</sup> He obtained the VDW radii of these ions based on the VDW radii of noble-gas atoms and the quantum scaling principle (QMSP), which states that for ions and atoms which are isoelectronic, any characteristic radius associated with a certain electron is inversely proportional to the effective nuclear charge.<sup>12</sup> In 1967, Bader et al. demonstrated that an outer contour of the electron density of 0.002 au could be used to define the VDW envelope.<sup>19</sup> In 1977, Boyd determined the atomic sizes of H to Xe by empirically scaling the radii of the 0.0001 au contour.<sup>20</sup> In the following work, rather than using radii which were then scaled back, Bader et al. determined the atomic and molecular volumes based on 0.001 and 0.002 au criteria directly.<sup>21</sup> Afterward, Rahm et al. proposed that 0.001 au is an appropriate density cutoff for determining VDW radii, and they determined the VDW radii of atoms, monovalent charged cations, and monovalent anions of elements 1–96.<sup>22</sup> Their predictions agreed well with the VDW radii of noble-gas atoms that were determined by Vogt and Alvarez.<sup>23</sup> They noted that this method can complement experimental data, especially when experimental data is not available.

The 12-6 Lennard-Jones (LJ) nonbonded model is widely used to simulate ions. It consists of Coulomb and LJ terms representing the electrostatic and VDW interactions, respectively. Our previous study indicated that one could find numerous parameters in the 12-6 LJ model to reproduce the same physical property.<sup>24</sup> To solve this issue, we fitted a curve based on the VDW parameters of noble-gas atoms to correlate the  $R_{\min}/2$  and  $\epsilon$  values in the LJ equation.<sup>24</sup> Using this noble-gas curve (NGC), we parameterized the LJ parameters for various ions.<sup>24–26</sup> The parameters agreed well with the VDW radii calculated based on the QMSP method and showed excellent transferability to other systems such as metalloproteins and salt solutions.<sup>24–26</sup> In addition, we found that for atomic ions with a charge of +2 or higher, one cannot determine a set of 12-6 LJ parameters that are in a physical range and can reproduce the hydration free energy (HFE) and ion–oxygen distance (IOD) values simultaneously.<sup>24–26</sup> And we attributed this to overlooking the ion-induced dipole interaction in the 12-6 nonbonded model, based on which we developed the 12-6-4 model to solve this issue.<sup>27</sup> The 12-6-4 model showed significant improvement over the 12-6 model: it could well reproduce both properties at the same time and showed excellent transferability to other systems.<sup>25–27</sup>

Recently, Zhang et al. designed a new set of VDW parameters for the 12-6 model (z12-6) to reproduce the HFE and IOD values at the same time.<sup>28–30</sup> However, the  $R_{\min}/2$  parameters they obtained were quite close to the Shannon ionic radii, which are significantly smaller than the VDW radii. The sum of the Shannon ionic radii gives the distance between counterions in crystal structures, which are determined by not only the VDW interactions but also the strong electrostatic attractions between the counterions. Moreover, the  $\epsilon$  values they obtained were quite large (e.g., can be thousands of kcal/mol for highly charged metal ions), which is contraindicated given the nature of the dispersion interaction. Hence, although their parameters were able to simultaneously reproduce the HFE and IOD values for an ion with a charge of +2 or higher, we believe their parameters result from overfitting and could cause significant issues in molecular modeling.

In the present study, we tested different schemes to derive the VDW radii of atoms and ions based on the electron density distributions. Our results suggested that atomic radii determined based on the 0.0015 au electron iso-surface give an excellent agreement with the experimentally determined VDW radii for noble-gas atoms. Based on this, we calculated the VDW radii of various atoms and ions, and these results showed excellent agreement with our previously determined LJ parameters for atomic ions. However, these calculated VDW radii deviated significantly from the VDW parameter set from Zhang et al.<sup>28–30</sup> In addition, molecular dynamics (MD) simulations show that the Zhang et al. parameter set introduces significant artifacts when simulating ion-ligand and metalloprotein systems. In comparison, our 12-6-4 model showed significantly better performance for simulating these systems. Hence, the Zhang et al. parameters should be used with significant care because of serious transferability issues.

In sum, through quantum calculations, this study provides a consistent set of VDW radii for ions across the periodic table for their most common oxidation states and different spin states, which has not been reported to the best of our knowledge. Moreover, our study also indicated that physical VDW radii are critical for obtaining meaningful results in MD

simulations of ion-containing systems, while unphysical/overfitted parameters have limited transferability and should be used with caution.

## COMPUTATIONAL METHODS

**Quantum Mechanical Calculations.** In the present study, the quantum calculations were carried out using the Gaussian16 program (version C.01).<sup>31</sup> Afterward, the VDW radii of atoms and ions were obtained based on the wavefunction analysis using the Multiwfn program (version 3.7).<sup>32</sup> Following the previous work done by Rahm et al.,<sup>22</sup> we assume atoms and ions are spheres. Specifically, the radius of a particle was computed based on the contour covered volume with  $r = \sqrt[3]{3V/4\pi}$ , which was defined as the region encompassed by a specific electron density cutoff.<sup>22</sup> The Monte Carlo method was used for calculating the contour covered volume, which incorporated randomness for the calculations. In the Monte Carlo procedure,  $N = 100 \times 2^9$  particles were randomly distributed in a box. The  $k$  factor was set to 1.7 to define the box. These are the recommended values from the manual. Further details of the procedure and these parameters can be found in the Multiwfn manual. Our tests indicated that the uncertainty (one  $\sigma$  value of the Gaussian distribution) of the calculated radii is around or within 0.01 Å.

**12-6 and 12-6-4 Models.** The 12-6 LJ and 12-6-4 LJ nonbonded models for ions are employed in the present study. The potential equations between the metal ions and other particles in the 12-6 model (eq 1) and the 12-6-4 model (eq 2) are shown below.

$$\begin{aligned}
 U_{12-6}(r_{ij}) &= \frac{e^2 Q_i Q_j}{r_{ij}} + \frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^6} \\
 &= \frac{e^2 Q_i Q_j}{r_{ij}} + \epsilon_{ij} \left[ \left( \frac{R_{\min,ij}}{r_{ij}} \right)^{12} - 2 \left( \frac{R_{\min,ij}}{r_{ij}} \right)^6 \right] \\
 &= \frac{e^2 Q_i Q_j}{r_{ij}} + 4\epsilon_{ij} \left[ \left( \frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left( \frac{\sigma_{ij}}{r_{ij}} \right)^6 \right]
 \end{aligned} \quad (1)$$

$$\begin{aligned}
 U_{12-6-4}(r_{ij}) &= \frac{e^2 Q_i Q_j}{r_{ij}} + \frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^6} - \frac{C_4^{ij}}{r_{ij}^4} \\
 &= \frac{e^2 Q_i Q_j}{r_{ij}} + \epsilon_{ij} \left[ \left( \frac{R_{\min,ij}}{r_{ij}} \right)^{12} - 2 \left( \frac{R_{\min,ij}}{r_{ij}} \right)^6 \right] \\
 &\quad - \frac{C_4^{ij}}{r_{ij}^4} \\
 &= \frac{e^2 Q_i Q_j}{r_{ij}} + 4\epsilon_{ij} \left[ \left( \frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left( \frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] - \frac{C_4^{ij}}{r_{ij}^4}
 \end{aligned} \quad (2)$$

Herein, the first term is the electrostatic term, where  $e$  is the proton charge;  $Q_i$  and  $Q_j$  represent the partial charges of the atoms  $i$  and  $j$ , respectively, with the partial charge of the atomic ion in its oxidation state; and  $r_{ij}$  is the distance between atoms  $i$  and  $j$ . In terms of the VDW term,  $R_{\min,ij}$  is the distance where

the VDW interaction between atoms  $i$  and  $j$  has an energy minimum.  $\sigma_{ij}$  is the distance where the VDW interaction between these two atoms equals zero.  $\epsilon_{ij}$  is the well depth of the VDW interaction. For the VDW term, the Lorentz–Berthelot combining rules (eqs 3 and 4) were used, which is consistent with the AMBER<sup>33</sup> and CHARMM<sup>34</sup> force fields, as well as our previous studies<sup>24–27,35–37</sup> and the studies of Zhang et al.<sup>28–30</sup>

$$R_{\min,ij} = \frac{R_{\min,ii} + R_{\min,jj}}{2} = R_{\min,i} + R_{\min,j} \quad (3)$$

$$\epsilon_{ij} = \sqrt{\epsilon_i \times \epsilon_j} \quad (4)$$

Herein,  $R_{\min,ii}$  and  $R_{\min,jj}$  are the VDW diameters of atoms  $i$  and  $j$ , while  $R_{\min,i}$  and  $R_{\min,j}$  are the VDW radii of atom  $i$  and  $j$ . The difference between the 12-6 and 12-6-4 models is that the 12-6-4 model has an extra  $r^4$  term which represents the ion-induced dipole interactions.<sup>25–27</sup> In the present study, the  $C_4$  term between a divalent ion and the water oxygen in the OPC water model<sup>38</sup> (represented by  $C_4^{m-w}$ ) was taken from our previous study.<sup>35</sup> Physically,  $C_4$  is proportional to the polarizability of the particle which is interacting with the metal ion.<sup>27</sup> Hence, the  $C_4$  terms between the ion and atoms which are not water–oxygen (denoted as  $l$ ) are calculated using eq 5. Herein,  $\alpha_w$  is the polarizability of the water oxygen, while the polarizability of  $l$  is represented by  $\alpha_l$ .

$$C_4^{m-l} = C_4^{m-w} \times \frac{\alpha_l}{\alpha_w} \quad (5)$$

If the system has multiple ions, the  $C_4$  term between every two ions (denoted as  $i$  and  $j$ ) is calculated using eq 6. Herein,  $\alpha_i$  and  $\alpha_j$  are the polarizabilities of ions  $i$  and  $j$ , respectively.

$$C_4^{i-j} = C_4^{i-w} \times \frac{\alpha_j}{\alpha_w} + C_4^{j-w} \times \frac{\alpha_i}{\alpha_w} \quad (6)$$

In the present study, the polarizabilities of atoms and ions are from the \$AMBERHOME/dat/leap/parm/lj\_1264\_pol.dat file in the AMBER 2021 software package.<sup>39</sup>

**Molecular Dynamics Simulations.** In terms of simulating the ion–imidazole and ion–acetate systems in the aqueous phase, the OPC water model<sup>38</sup> was used to model the solvent box. Force field parameters of the imidazole and acetate are given in Figure S1 and Table S1, which are the same as in the previous study.<sup>40</sup> The AMBER ff14SB force field<sup>41</sup> was used to model the protein, imidazole, and acetate, with the additional parameters for imidazole listed in Table SI. In terms of the simulations of the metalloprotein systems, the PDB entry 1F9Z<sup>42</sup> was used for modeling the Ni<sup>2+</sup>-containing glyoxalase Glx I and the PDB entry 2YRH<sup>43</sup> was employed for modeling the zinc finger domain. For these simulations, each system was minimized, heated, equilibrated, and followed by a 100 ns production simulation. In these simulations, both the Zhang et al. 12-6 parameter set for Zn<sup>2+</sup> and Ni<sup>2+</sup> in conjunction with the OPC water model<sup>30</sup> and the Li/Merz 12-6-4 parameter set for Zn<sup>2+</sup> and Ni<sup>2+</sup> in conjunction with the OPC water model<sup>35</sup> were investigated.

## RESULTS AND DISCUSSION

In 2016, Rahm et al. determined the radii of atoms and ions of elements 1–96 based on the 0.001 au electron contour.<sup>22</sup> Specifically, they determined the radii for atoms and the corresponding singly charged cations and anions. However,



## VDW Radii of Ions

|                      |                      |                      |                                  |                                             |                                              |                                              |                                              |                                  |                                  |                                  |                                  |                      |                                  |                                              |                                  |                                  |                       |
|----------------------|----------------------|----------------------|----------------------------------|---------------------------------------------|----------------------------------------------|----------------------------------------------|----------------------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------|----------------------------------|----------------------------------------------|----------------------------------|----------------------------------|-----------------------|
| H<br>-1: 1.67        |                      |                      |                                  |                                             |                                              |                                              |                                              |                                  |                                  |                                  |                                  | 2<br>He              |                                  |                                              |                                  |                                  |                       |
| 3<br>Li<br>+1: 0.93  | 4<br>Be<br>+2: 0.74  |                      |                                  |                                             |                                              |                                              |                                              |                                  |                                  |                                  |                                  | 5<br>B<br>+3: 0.63   | 6<br>C<br>-4: 2.50<br>+4: 0.53   | 7<br>N<br>-3: 2.12                           | 8<br>O<br>-2: 1.87               | 9<br>F<br>-1: 1.65               | 10<br>Ne              |
| 11<br>Na<br>+1: 1.27 | 12<br>Mg<br>+2: 1.12 |                      |                                  |                                             |                                              |                                              |                                              |                                  |                                  |                                  |                                  | 13<br>Al<br>+3: 1.00 | 14<br>Si<br>-4: 2.78<br>+4: 0.92 | 15<br>P<br>-3: 2.49<br>+3: 1.56<br>+5: 0.84  | 16<br>S<br>-2: 2.27              | 17<br>Cl<br>-1: 2.06             | 18<br>Ar              |
| 19<br>K<br>+1: 1.67  | 20<br>Ca<br>+2: 1.52 | 21<br>Sc<br>+3: 1.40 | 22<br>Ti<br>+3: 1.38<br>+4: 1.28 | 23<br>V<br>+2: 1.56<br>+3: 1.41<br>+4: 1.31 | 24<br>Cr<br>+2: 1.58<br>+3: 1.38<br>+6: 1.15 | 25<br>Mn<br>+2: 1.56<br>+4: 1.25<br>+7: 1.08 | 26<br>Fe<br>+2: 1.56<br>+3: 1.38<br>+5: 1.16 | 27<br>Co<br>+2: 1.54<br>+3: 1.38 | 28<br>Ni<br>+2: 1.51             | 29<br>Cu<br>+1: 1.65<br>+2: 1.50 | 30<br>Zn<br>+2: 1.46             | 31<br>Ga<br>+3: 1.32 | 32<br>Ge<br>+4: 1.22             | 33<br>As<br>-3: 2.71<br>+3: 1.72<br>+5: 1.14 | 34<br>Se<br>-2: 2.63             | 35<br>Br<br>-1: 2.27             | 36<br>Kr              |
| 37<br>Rb<br>+1: 1.84 | 38<br>Sr<br>+2: 1.71 | 39<br>Y<br>+3: 1.59  | 40<br>Zr<br>+4: 1.48             | 41<br>Nb<br>+3: 1.65<br>+5: 1.38            | 42<br>Mo<br>+6: 1.34                         | 43<br>Tc<br>+6: 1.36                         | 44<br>Ru<br>+3: 1.61                         | 45<br>Rh<br>+3: 1.60             | 46<br>Pd<br>+2: 1.71<br>+4: 1.47 | 47<br>Ag<br>+1: 1.82<br>+2: 1.67 | 48<br>Cd<br>+2: 1.64             | 49<br>In<br>+3: 1.53 | 50<br>Sn<br>+4: 1.42             | 51<br>Sb<br>-3: 2.77<br>+3: 1.89             | 52<br>Te<br>-2: 2.75             | 53<br>I<br>-1: 2.45              | 54<br>Xe              |
| 55<br>Cs<br>+1: 2.04 | 56<br>Ba<br>+2: 1.91 | LA                   | 72<br>Hf<br>+4: 1.50             | 73<br>Ta<br>+5: 1.42                        | 74<br>W<br>+6: 1.35                          | 75<br>Re<br>+3: 1.70<br>+4: 1.61<br>+7: 1.30 | 76<br>Os<br>+3: 1.69<br>+4: 1.60<br>+6: 1.46 | 77<br>Ir<br>+3: 1.68             | 78<br>Pt<br>+2: 1.80             | 79<br>Au<br>+3: 1.65             | 80<br>Hg<br>+1: 1.97<br>+2: 1.76 | 81<br>Tl<br>+3: 1.64 | 82<br>Pb<br>+4: 1.54             | 83<br>Bi<br>+3: 1.78                         | 84<br>Po<br>+2: 2.01<br>+4: 1.69 | 85<br>At<br>-1: 2.51             | 86<br>Rn              |
| 87<br>Fr<br>+1: 2.13 | 88<br>Ra<br>+2: 1.98 | AC                   | 104<br>Rf                        | 105<br>Db                                   | 106<br>Sg                                    | 107<br>Bh                                    | 108<br>Hs                                    | 109<br>Mt                        | 110<br>Ds                        | 111<br>Rg                        | 112<br>Cn                        | 113<br>Nh            | 114<br>Fl                        | 115<br>Mc                                    | 116<br>Lv                        | 117<br>Ts                        | 118<br>Og             |
|                      |                      |                      | 57<br>La<br>+3: 1.80             | 58<br>Ce<br>+3: 1.78                        | 59<br>Pr<br>+3: 1.75                         | 60<br>Nd<br>+3: 1.73<br>+4: 1.65             | 61<br>Pm<br>+3: 1.71                         | 62<br>Sm<br>+2: 1.78<br>+3: 1.70 | 63<br>Eu<br>+2: 1.77<br>+3: 1.68 | 64<br>Gd<br>+3: 1.65             | 65<br>Tb<br>+3: 1.65<br>+4: 1.58 | 66<br>Dy<br>+3: 1.64 | 67<br>Ho<br>+3: 1.60             | 68<br>Er<br>+3: 1.55                         | 69<br>Tm<br>+3: 1.57             | 70<br>Yb<br>+2: 1.65<br>+3: 1.55 | 71<br>Lu<br>+3: 1.56  |
|                      |                      |                      | 89<br>Ac<br>+3: 1.88             | 90<br>Th<br>+4: 1.79                        | 91<br>Pa<br>+5: 1.71                         | 92<br>U<br>+3: 1.82<br>+6: 1.64              | 93<br>Np<br>+3: 1.82                         | 94<br>Pu<br>+4: 1.70             | 95<br>Am<br>+3: 1.77             | 96<br>Cm<br>+3: 1.75             | 97<br>Bk<br>+3: 1.72             | 98<br>Cf<br>+3: 1.72 | 99<br>Es<br>+3: 1.71             | 100<br>Fm<br>+3: 1.71                        | 101<br>Md<br>+3: 1.69            | 102<br>No<br>+3: 1.67            | 103<br>Lr<br>+3: 1.64 |

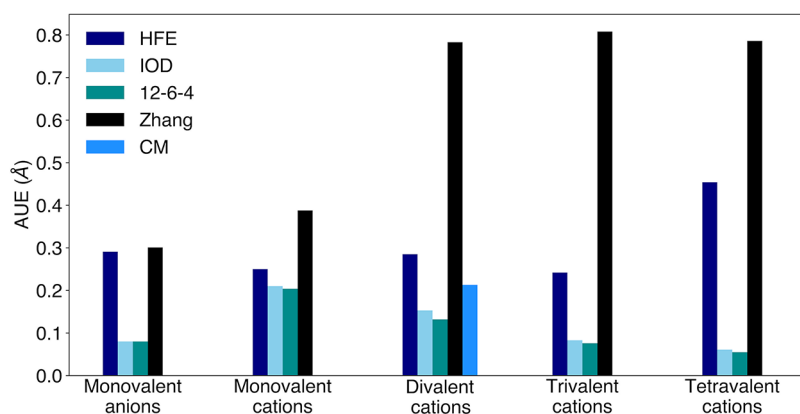
**Figure 1.** Calculated VDW radii of select ions in the periodic table. Only the VDW radii of ions with the most common oxidation states are shown, with the oxidation state specified before the radius. For a certain oxidation state, multiple spin states can be available, herein only the radius of the lowest spin state for each oxidation state is shown. Calculated VDW radii of other species can be seen in [Tables S5 and S6](#).

many ions have oxidation states larger than +1 or more negative than −1. In the present study, following their work, we have determined the van der Waals (VDW) radii for various elements across the periodic table and their associated ions based on the electron density contour. Specifically, in our study, we found that the 0.0015 au contour would be a better criterion for evaluating the VDW volumes/radii, and we have calculated the VDW radii for atoms and ions with different oxidation states and spin states based on this criterion. This work provides a systematic set of the VDW radii of atomic cations and anions.

To test the sensitivity of the calculated radii to the levels of theory, we benchmarked the results by varying the method ([Table S2](#)) and the basis set ([Table S3](#)), respectively. In these benchmarks, we used the electron density of 0.0015 au as the cutoff because it showed the best agreement compared to the experimentally determined VDW radii of noble-gas atoms (discussed below). In particular, we used the Hartree–Fock,<sup>44</sup> B3LYP,<sup>45,46</sup> PBE0,<sup>47,48</sup> MP2,<sup>49</sup> and MP4SDQ<sup>50</sup> methods along with the aug-cc-pVTZ basis set<sup>51,52</sup> in our tests ([Table S2A](#)). Our results showed that these methods provided similar radii when using the same basis set. Although the Hartree–Fock method does not include electron correlation, it showed similar results to other levels of theory, which agrees with previous work.<sup>22</sup> Compared to cations, the electron cloud of anions is more dispersed, causing their wavefunctions to be more likely dependent on the density functional. Because of this reason, we also benchmarked different density functionals for calculating the VDW radii of anions ([Table S2B](#)). Specifically, the B3LYP,<sup>45,46</sup> PBE0,<sup>47,48</sup> BLYP,<sup>46,53</sup> and PBE<sup>48</sup> functionals were tested in conjugation with the SDD (Stuttgart/Dresden effective core potential (ECP)) basis set.<sup>31</sup> The results indicated that the VDW radii calculated in the present way are not particularly sensitive to the functional choice. This may be because we are dealing with atomic

anions, which are very small systems. Finally, the B3LYP density functional was chosen because it has been one of the most widely used functionals due to its excellent performance and relatively low computational cost. In addition, we benchmarked the results by B3LYP with various basis sets, ranging from the Pople basis sets to the correlation-consistent basis sets and to the SDD basis set ([Table S3](#)). We also found that the results were relatively insensitive to the basis set chosen. Again, this may be because the investigated systems (atoms and atomic ions) are very small, hence the basis set effect is also small. Finally, we chose the B3LYP/SDD level of theory to carry out the calculations because the SDD basis set<sup>31</sup> covers almost the entire periodic table and estimates the relativistic effect for heavy atoms (using ECPs on the atoms beyond Ar).

As mentioned above, both 0.001 and 0.002 au have been used as the criteria previously to determine the VDW volumes of atoms and molecules. However, the criterion 0.0015 au has not been used before. Herein, we benchmarked all of these three criteria for calculating the VDW radii of noble-gas atoms (He to Xe), and the results are shown in [Table S4](#). We observe that the 0.0015 au set showed the smallest average unsigned errors (AUEs) toward the five of six data sets that were determined experimentally. In comparison, the 0.001 au provided significantly larger VDW radii, yielding AUEs consistently higher compared to these experimental data sets. Meanwhile, 0.002 au showed significantly smaller VDW radii, also providing higher AUEs toward the experimental data sets in general. The data set of Rahm et al.<sup>22</sup> was obtained based on the 0.001 au criterion, yielding the VDW radii of He, Ne, Ar, Kr, and Xe as 1.34, 1.56, 1.97, 2.12, and 2.32 Å, respectively, which are very close to the 0.001 au set we obtained herein (with an average unsigned difference of ~0.014 Å), although they used a different level of theory (PBE0/ANO-RCC). These results agree with our above calculations that the



**Figure 2.** Average unsigned errors (AUEs) of the VDW radii in the Li/Merz parameter sets (HFE, IOD, CM, 12-6-4) and the Zhang et al. parameter set for atomic ions with different charges in conjunction with the OPC water model. These AUEs were calculated using the calculated VDW radii in the present study as the reference.

calculated VDW radii are insensitive to the level of theory one uses.

Using the 0.0015 au criterion, we have determined the VDW radii for atoms and atomic ions across the periodic table. Specifically, we have calculated the VDW radii for various atoms and their atomic ions with the most common oxidation states. These results are provided in Tables S5 and S6, with select VDW radii of ions being shown in Figure 1. Note that for some of the ions (e.g.,  $\text{Ga}^{3+}$ ,  $\text{Ge}^{4+}$ ) the SDD basis set does not have any electrons described explicitly (i.e. all of the electrons in these ions are described by the effective core potentials); hence, we calculated their VDW radii based on the Def2SV basis set<sup>54,55</sup> and then scaled these radii based on a factor which was obtained by comparing the calculated radii using the SDD and Def2SV basis sets for the neutral atoms of these ions (see Table S7).

**Comparison with the VDW Radii Calculated Based on the Quantum Scaling Principle (QMSP).** Stokes calculated the VDW radii of various cations and anions in 1964.<sup>12</sup> He obtained the VDW radii of noble-gas atoms based on different experiments, and then he estimated the VDW radii of cations and anions which are isoelectronic to noble-gas atoms based on the QMSP, that is “within an isoelectronic sequence of ions or atoms, any characteristic radius  $r'$  associated with a given electron is inversely proportional to the effective atomic number ( $Z-S$ ), where the screening parameter  $S$  is constant for the isoelectronic sequence.”<sup>12</sup> Because Stokes did not discuss the origins of this principle in the 1964 paper, we hypothesize that its origin comes from the hydrogenic wavefunctions which has the form of  $\psi \propto e^{-Z_{\text{eff}}r/na_0}$ , where  $Z_{\text{eff}} = Z - S$ , and  $n$  is the principal quantum number. However, we note that this is just an approximation, where the  $Z_{\text{eff}}$  term is not only in the exponential term but also in the normalization factor (the 1s orbital is shown in eq 7 as an example).

$$\psi(1s) = \frac{1}{\sqrt{\pi}} \left( \frac{Z_{\text{eff}}}{a_0} \right)^{3/2} e^{-Z_{\text{eff}}r/na_0} \quad (7)$$

However, the QMSP method does not consider the effect of  $Z_{\text{eff}}$  in the normalization factor, which causes the QMSP values to deviate from our calculated values (Table S8). Such a deviation is minimal for the monovalent cations. However, this deviation is significantly larger for anions than cations, and it becomes larger when the ion charge increases.

**Comparison with the Shannon Ionic Radii.** In the studies of Zhang et al.,<sup>28–30</sup> they noted that their  $R_{\text{min}}/2$  parameters agreed well with the Shannon ionic radii.<sup>4</sup> We have compared the calculated VDW radii in the present study with the Shannon ionic radii (Table S8). We can see that the calculated VDW radii are significantly larger than the ionic radii for the same species. This agrees with our discussion above that the ionic radii are determined by not only VDW interactions but also electrostatic interactions; hence, the ionic radii should be significantly smaller than the VDW radii that were determined only by the VDW interactions. This principle should be always valid except in the case of noble-gas atoms, for which the ionic radii and VDW radii would agree with each other because only VDW interactions are available between these atoms.

**Comparison with the VDW Radii of Ions in Different Parameter Sets.** To make a direct comparison between the z12-6 parameter set and the calculated VDW radii in the present study, we give these values in Table S9, along with the VDW radii from the Li/Merz parameter sets. Herein, the ion parameters for the OPC water model were used as an example, while the ion parameters for other water models follow the same trends. By referring to the calculated VDW radii in the present study as the standard, the average unsigned errors (AUEs) of different parameter sets are shown in Figure 2. From Table S9 and Figure 2, we can see that the z12-6 parameters are significantly smaller than the calculated VDW radii, with the AUE increasing along with the ion charge. This means that the higher the charge of the ion, the larger the error of the VDW radii used by the z12-6 parameter set. For the tetravalent metal ions, the AUE can be as high as  $\sim 0.97$  Å. In comparison, the Li/Merz parameter sets showed much smaller AUEs for these ions, with the IOD and 12-6-4 parameter sets providing the lowest AUEs. The above comparison indicates that reproduction of both the HFE and IOD values for an atomic ion at the same time can lead to overfitting, which does not ensure a high-quality model. Such an observation is consistent with that in z12-6 parameters set where the  $\epsilon$  value increases along with the ion charge. For example, their parameter set for the OPC water model has  $\epsilon$  values (i.e., the well depth or interaction strength) for  $\text{Zn}^{2+}$ ,  $\text{Fe}^{3+}$ , and  $\text{Hf}^{4+}$  of 225.8396, 1655.6733, and 3388.4415 kcal/mol, respectively. These values are inconsistent with the well-known weakness of VDW interactions.

**Transferability Issue.** In both the Li/Merz parameter sets and the Zhang et al. parameter set, the VDW parameters follow the Lorentz–Berthelot combining rules. According to these combining rules, the equilibrium distance of the VDW interaction is equal to the sum of the VDW radii of the two interacting particles:  $R_{\min,ij} = R_{\min,i} + R_{\min,j}$ , which is based on the hard-sphere approximation, while the well depth of the VDW interaction is equal to the geometric average of the  $\epsilon$  values of the two interacting particles  $\epsilon_{ij} = \sqrt{\epsilon_i \times \epsilon_j}$ . Because Zhang et al.'s parameters are unbalanced (i.e.,  $R_{\min}/2$  is too small while  $\epsilon$  is too large), we expect them to have transferability issues from water to more complex systems.

To characterize transferability, we use  $\text{Zn}^{2+}$  and  $\text{Hf}^{4+}$  as examples, where we compared the z12-6 parameter set with our 12-6-4 parameter set for generating the VDW parameters for different interaction pairs between metal ions and ligating atoms (Table S10). These ligating atoms are the oxygen atom from water, the carboxylate oxygen from ASP/GLU, the  $\delta/\epsilon$  nitrogen from HID/HIE, and sulfur from CYM, which represent the most common coordinating atoms in metalloproteins. For both  $\text{Zn}^{2+}$  and  $\text{Hf}^{4+}$ , we found that the z12-6 parameters provide significantly larger  $A_{ij}$  and  $B_{ij}$  values than the Li/Merz 12-6-4 parameter set, which implies that these parameters would not have the same transferability to mixed systems as the Li/Merz parameters.

**Transferability to Ion–Ligand Systems in Aqueous Phase.** To investigate the transferability of the z12-6 parameters for modeling ion–ligand systems, we calculated the binding free energy between  $\text{Zn}^{2+}$  and imidazole in a water box. For comparison, we also calculated the binding free energy between  $\text{Zn}^{2+}$  and imidazole in the aqueous phase based on the 12-6-4 parameter set for  $\text{Zn}^{2+}$  (the default 12-6-4). Specifically, we performed potential of mean force (PMF) calculations between  $\text{Zn}^{2+}$  and the ligating nitrogen atom of imidazole, based on which we determined the binding free energy between  $\text{Zn}^{2+}$  and imidazole in the aqueous phase. The simulated results are shown in Table 1, together with the

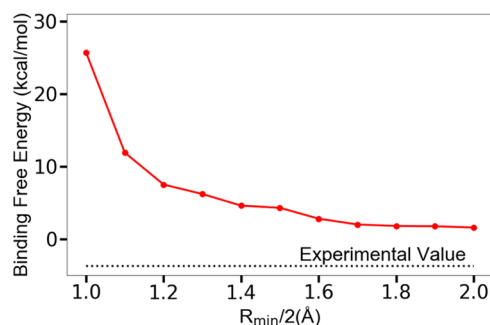
**Table 1.** List of Binding Free Energies between  $\text{Zn}^{2+}$  and Imidazole and Acetate in Aqueous Phase

| parameter set       | imidazole<br>(kcal/mol) | acetate monodentate<br>(kcal/mol) | acetate bidentate<br>(kcal/mol) |
|---------------------|-------------------------|-----------------------------------|---------------------------------|
| default 12-6-4      | 3.14                    | 0.14                              | 1.96                            |
| Zhang et al.        | 17.73                   | −11.60                            | −14.48                          |
| experimental values | −3.7 <sup>a</sup>       | −1.2 <sup>b</sup>                 | >−1.2 <sup>c</sup>              |

<sup>a</sup>Based on the log  $K$  of 2.663 from ref 56. <sup>b</sup>Based on the log  $K$  of 0.91 from ref 57. <sup>c</sup>This value was determined based on that monodentate is preferred for  $\text{Zn}^{2+}$  with acetate in aqueous solution.<sup>58</sup>

experimental value. We can see that the z12-6 parameters significantly underestimated the binding affinity between  $\text{Zn}^{2+}$  and imidazole by  $\sim 20$  kcal/mol. In comparison, although the 12-6-4 parameter set also underestimates this binding free energy, the underestimation is  $\sim 7$  kcal/mol, which is only one-third of the error of the z12-6 parameter set. Furthermore, in a previous study, it was shown that the ion–imidazole binding affinity can be accurately reproduced by optimizing the  $C_4$  term in the 12-6-4 model.<sup>40</sup> Hence, because the z12-6 parameters were overfitted, with  $R_{\min}/2$  too small and  $\epsilon$  too large, it has limited transferability to describe  $\text{Zn}^{2+}$ –imidazole interactions.

To further investigate the performance of the 12-6 model for describing the divalent ion–imidazole interactions in aqueous phase, we performed a parameter scan. Specifically, to keep the  $R_{\min}/2$  and  $\epsilon$  values in physical ranges, we generated the  $\epsilon$  parameter based on the  $R_{\min}/2$  value using the NGC. Finally, we plotted our results in Figure 3, which illustrates a curve of



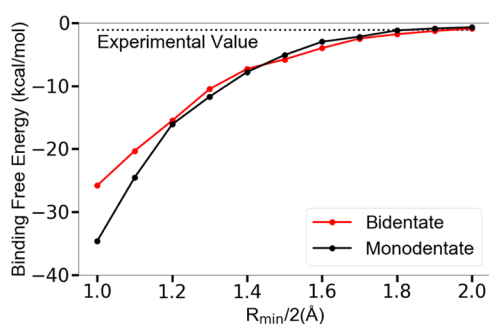
**Figure 3.** Parameter scan results of the binding affinity between a divalent metal ion and imidazole in an OPC water box using the 12-6 model.

the  $R_{\min}/2$  values and the simulated binding affinities. Figure 3 indicates that as the VDW radius increases, the divalent ion–imidazole binding affinity becomes less positive. However, when  $R_{\min}/2$  is in the physically meaningful range, it is impossible to reproduce the binding affinity between  $\text{Zn}^{2+}$  and imidazole based on the current protocol (HID charges on imidazole and the OPC water model), indicating the limitations of the 12-6 model to simulate ion–ligand interactions in aqueous phase. Again, in comparison, it was shown that the ion–imidazole binding affinity can be accurately reproduced by optimizing the  $C_4$  term in the 12-6-4 model.<sup>40</sup>

Acetate is another common ligand for metal ions in metalloproteins. Herein, we also calculated the binding free energy between  $\text{Zn}^{2+}$  and acetate using the z12-6 parameter set and the 12-6-4 model, and the results are shown in Table 1 as well. Unlike the  $\text{Zn}^{2+}$ –imidazole interaction, the z12-6 parameter set significantly overestimates the  $\text{Zn}^{2+}$ –acetate interaction ( $>10$  kcal/mol) and incorrectly predicts that the bidentate mode is more favorable than the monodentate one. All of these results indicate that there is a significant overestimation of the  $\text{Zn}^{2+}$ –carboxylate interactions when using the z12-6 parameter set. In comparison, the 12-6-4 parameters correctly predicted the preference of the monodentate mode over the bidentate mode and showed much better agreement in terms of the binding affinity (off only by  $\sim 1.3$  kcal/mol). Furthermore, a previous study has shown that the 12-6-4 model can model the chelate effect after optimizing the  $C_4$  parameter.<sup>59</sup>

Moreover, we also performed the parameter scan for a divalent metal ion with acetate. In this scan, the OPC water model was used as well, and the acetate partial charges are shown in Table S1. Again, the NGC was used to restrain the relationship between  $R_{\min}/2$  and  $\epsilon$ . The scan results are shown in Figure 4. Unlike  $\text{Zn}^{2+}$ –imidazole, we can see that it is possible to adjust the  $R_{\min}/2$  parameter to reproduce the experimental value, which is  $-1.2$  kcal/mol for the  $\text{Zn}^{2+}$ –acetate binding affinity in aqueous phase.<sup>57</sup> However, such an  $R_{\min}/2$  value ( $\sim 1.8$  Å) is significantly larger than the VDW radius of  $\text{Zn}^{2+}$  (calculated as 1.456 Å in the present study). In





**Figure 4.** Parameter scan results of the binding affinity between a divalent metal ion and acetate in an OPC water box using the 12-6 model. Herein, two curves are shown, one is for the monodentate form and the other one is for the bidentate form. The binding affinities for different binding modes were obtained based on the PMF plot between  $\text{Zn}^{2+}$  and the carboxylate carbon of the acetate in a water box.

addition, while reproducing the binding free energy with acetate, the bidentate mode is slightly more favored, which is in contrast to the experimental result where the monodentate form is more favored by  $\text{Zn}^{2+}$ .<sup>58</sup> Again, in comparison, we showed that the 12-6-4 model can reproduce the coordination mode between a divalent metal ion and acetate through refinement of the  $\text{C}_4$  parameter.<sup>40</sup>

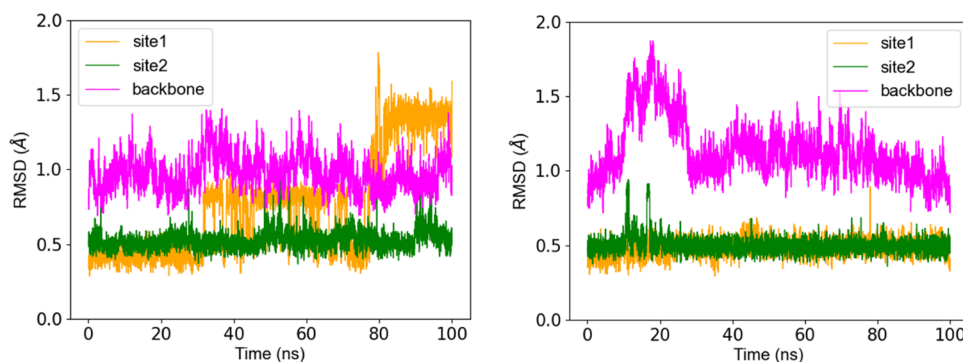
In sum, our calculations show that the z12-6 parameter set has serious transferability issues when simulating ion-ligand systems. Specifically, our simulations indicated that the z12-6 parameters underestimate the  $\text{Zn}^{2+}$ –imidazole interaction but overestimate the  $\text{Zn}^{2+}$ –acetate interaction, while the 12-6-4 parameters showed significant improvement for describing these interactions.

**Transferability to Metalloprotein Systems.** To further test the transferability of the z12-6 parameters to complex systems, we performed MD simulations for  $\text{Ni}^{2+}$ -containing glyoxalase Glx I (PDB ID: 1F9Z)<sup>42</sup> and a zinc finger domain from the zinc finger protein 473 (PDB ID: 2YRH) using the z12-6 parameter set and the 12-6-4 parameter set. Again, the OPC water model was used to simulate the solvent in these simulations.

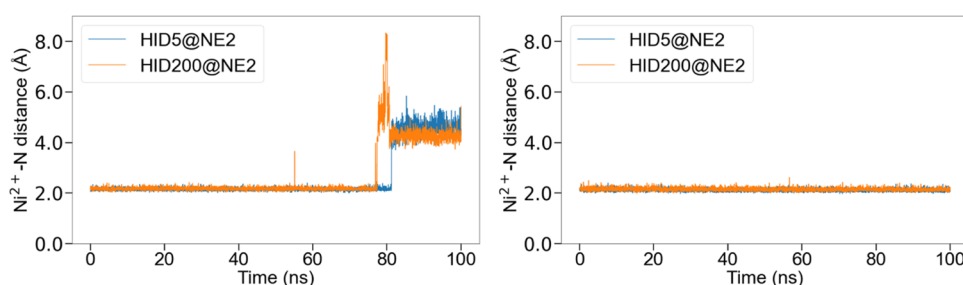
Figure 5 shows the RMSD values of two  $\text{Ni}^{2+}$  binding sites of Glx I (with site 1 consisting of HID5, GLU56, HID200, GLU248, and two water molecules, and site 2 consisting of HID74, GLU122, HID131, GLU182, and two water molecules) and the RMSD values of the protein backbone N,  $\text{C}_\alpha$ , and C atoms. By comparing the results based on the

z12-6 parameter set and the 12-6-4 parameter set, we found the latter successfully maintained the coordination modes of both metal sites during the simulations, not unlike what we observed previously for an  $\text{Fe}^{3+}$  containing metalloprotein.<sup>36</sup> In contrast, the z12-6 parameters could not maintain the structure of metal site 1, showing an RMSD of  $\sim 1.4$  Å after 80 ns. To better understand this, a distance analysis was carried out. We found both HID5 and HID200 lost their coordination to  $\text{Ni}^{2+}$  around 80 ns (Figure 6). To better illustrate these structural changes, we depict the last snapshot of site 1 in glyoxalase Glx I from the MD simulations in Figure 7. We can see that the z12-6 parameter set loses the two HID residues complexed to  $\text{Ni}^{2+}$ , while both Glu56 and Glu248 are bidentate to  $\text{Ni}^{2+}$ . This is consistent with our PMF simulations, which showed that the z12-6 parameter set underestimates the  $\text{Zn}^{2+}$ –imidazole interaction but overestimates the  $\text{Zn}^{2+}$ –acetate interaction and prefers bidentate coordination over monodentate (although the PMF calculations were for  $\text{Zn}^{2+}$ , we would expect to have similar results for  $\text{Ni}^{2+}$ ). In comparison, the 12-6-4 parameter set successfully maintained the metal coordination mode with two monodentate GLU residues, two HID residues, and two water molecules over the MD simulations (Figures 5–7).

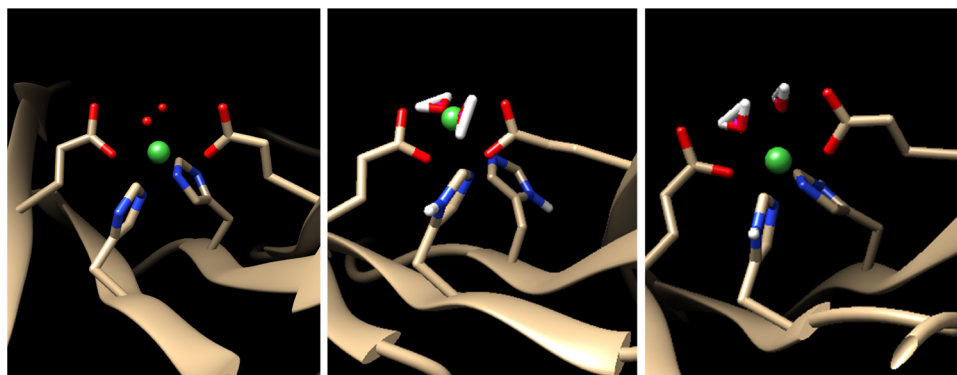
Figure S2 shows the RMSD values of the zinc-binding site (CYM13, CYM16, HID29, HID33) and of the protein backbone during the MD simulations using the z12-6 parameter set and the 12-6-4 parameter set. The 12-6-4 parameter set showed lower RMSD values of the zinc site compared to the z12-6 parameter set, with the latter showing an increasing RMSD value after  $\sim 10$  ns. A distance analysis was performed between  $\text{Zn}^{2+}$  and the two histidine residues (Figure S3). We found that the fluctuations of these Zn–N distances are significantly larger for the z12-6 parameter set than those of the 12-6-4 parameter set. Figure S4 shows the last snapshot of the metal site for each simulation. We observe that the z12-6 parameter set yielded a five-coordinate structure, with an additional water coming in to bind to  $\text{Zn}^{2+}$ . This agrees with our PMF simulations that the z12-6 parameter set underestimates the  $\text{Zn}^{2+}$ –imidazole interactions, which could cause water to out-compete the histidine residues in the metal coordination. In contrast, the 12-6-4 parameter set successfully maintained the zinc-binding mode formed by two CYM and two HID residues over the timescale of the MD simulations (Figures S2–S4), indicating its better transferability over the z12-6 parameter set.



**Figure 5.** RMSD values of the metal site residues (the six ligating residues of  $\text{Ni}^{2+}$ ) in the two  $\text{Ni}^{2+}$  binding sites and the protein backbone N,  $\text{C}_\alpha$ , and C atoms of the glyoxalase Glx I system. The left panel is for the z12-6 parameter set, while the right panel is for the 12-6-4 parameter set.



**Figure 6.** Distance between HID5@NE2 and  $\text{Ni}^{2+}$  or between HID200@NE2 and  $\text{Ni}^{2+}$  in the first metal binding site of Glx I. The left panel is for the z12-6 parameter set, while the right panel is for the 12-6-4 parameter set. The distorted binding pocket leads to enlarged  $\text{Ni}^{2+}$ –N distances when using the z12-6 parameter set.



**Figure 7.** (Left) Crystal structure of the  $\text{Ni}^{2+}$  binding site I, along with the last snapshot of this metal site from the MD simulations using (middle) the Zhang et al.'s parameter set or (right) the 12-6-4 parameter set.

Overall, our simulations indicate that even though the z12-6 parameter set reproduces both the experimental HFE and IOD for an ion, it has serious transferability issues caused by overfitting of the model into unphysical parameter space. Specifically, it has unbalanced ion–water and ion–ligand interactions and would likely yield distorted structures when simulating metalloprotein systems. Moreover, it is difficult for the 12-6 model, in general, to balance the ion–water and ion–ligand interactions even if using parameters in a more physical range. In comparison, the 12-6-4 model, which is based on physical consideration, has shown excellent performance in simulating ions in aqueous phase and well-behaved transferability to modeling metalloproteins. Previous work also showed with further refinement, the 12-6-4 model can be systematically improved.<sup>40,59</sup> Therefore, we would recommend the use of the 12-6-4 model instead of the 12-6 model to model metalloproteins as the former would give a more physical and accurate description along with better transferability.

## CONCLUSIONS

Atomic radii play fundamental roles in scientific research. There are different atomic radii available: covalent radii, ionic radii, VDW radii, etc. Although crystal structures have been used to derive the covalent radii and ionic radii of elements as well as the VDW radii of noble-gas atoms, it is not straightforward to derive the VDW radii of ions based on the crystal structures. This is because the electrostatic interaction and VDW interaction together determine the atomic distances in such a case, while decomposing them is not straightforward and usually involves assumptions. Determination of the VDW volume of an atom or a molecule based on

the electron density contour at a certain cutoff has been used in previous studies. In the present study, we found that using a 0.0015 au electron density cutoff, one can obtain the VDW radii of noble-gas atoms which have better agreement with experimentally determined values than using a 0.001 or 0.002 au cutoff. Based on this criterion, we have calculated the VDW radii of various atomic ions across the periodic table, thereby providing a consistent set of VDW radii of ions.

The 12-6 Lennard-Jones equation has been widely used to describe the VDW interactions in force fields. In our previous studies, we showed that the 12-6 nonbonded model of ions could not reproduce both the HFE and IOD values of an ion simultaneously when its VDW parameters are in physical range and its charge is +2 or higher.<sup>24–26</sup> To solve this issue, we developed the 12-6-4 model by adding an extra term to represent the ion-induced dipole interaction<sup>27</sup> and we showed that the 12-6-4 model could reproduce both the HFE and IOD at the same time, along with excellent transferability to complex systems such as metalloproteins.<sup>25–27</sup> However, recent studies from Zhang et al. showed that one could simultaneously reproduce the HFE and IOD for an ion with charge of +2 or higher by using the 12-6 model<sup>28–30</sup> and the VDW radii obtained by them were close to the Shannon ionic radii.<sup>4</sup> Herein, we showed that our VDW parameters have much better agreement with the QM-calculated VDW radii. In contrast, Zhang et al.'s parameter set were found to be overfitted and deviated significantly from these physically meaningful values.

In addition, we performed MD simulations based on the z12-6 parameter set and our 12-6-4 parameter set. Our results indicated the z12-6 parameter set showed large deviations when simulating the ion–imidazole and ion–acetate binding affinities. In comparison, the 12-6-4 parameter set showed



significantly improved performance. Moreover, the 12-6-4 parameter set retains the metal site structures when simulating the Ni<sup>2+</sup>-containing glyoxalase Glx I and a zinc finger domain. However, the z12-6 parameter set provided distorted structures for these metal sites. Furthermore, we found that in general it is difficult for the 12-6 model to describe ion–ligand interactions in the aqueous phase when using physically meaningful parameters. In comparison, the 12-6-4 model could solve this issue after parameter optimization.<sup>40,59</sup> Hence, our recommendation is the 12-6-4 model is preferred for highly charged ions. Indeed, this model gives better answers for the right reason. In contrast, for these systems, the 12-6 or z12-6 model should be used with caution when simulating complex systems such as metalloproteins because of the parameter transferability issue.

In summary, this study provides a consistent set of VDW radii for ions and showed that it is important to have the VDW radii of ions in a physically meaningful range when developing and using force field models for highly charged ions in MD simulations.

## ■ ASSOCIATED CONTENT

### SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jctc.2c01255>.

Force field parameters; calculated van der Waals radii for atoms and ions; and simulation results of the zinc finger domain (PDF)

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### Notes

The authors declare no competing financial interest.

## ■ ACKNOWLEDGMENTS

This work was partially supported by the start-up funds from Loyola University Chicago (to P.L.) and partially supported by the grants from the National Institutes of Health (to K.M.M.,

Grant Numbers GM044974 and GM066859). The authors thank Dr. Lin Song for providing the script for the potential-mean-force calculations. P.L., M.S., and L.L. acknowledge the computational support from the Extreme Science and Engineering Discovery Environment (XSEDE).<sup>60</sup> XSEDE was funded by the National Science Foundation (NSF) with Grant Number ACI-1548562. Specifically, this work used the computational resources on Expanse at the San Diego Supercomputer Center (Allocation Numbers TG-CHE200102 and TG-BIO210105). Z.L. and K.M.M. thank the computational support from the High Performance Computing Center (HPCC) in the Institute of Cyber-Enabled Research (iCER) at Michigan State University (MSU).

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